

The Internet



Most of it is garbage anyways

Your expectations

Please visit
<https://forms.office.com/e/XwSnZAkjwc>

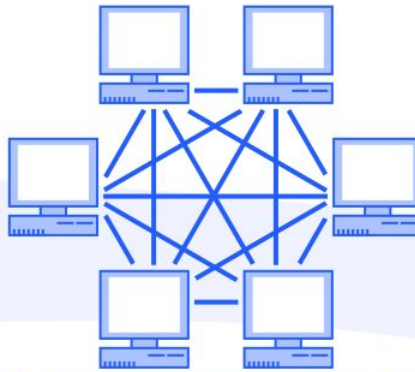


or use the QR code

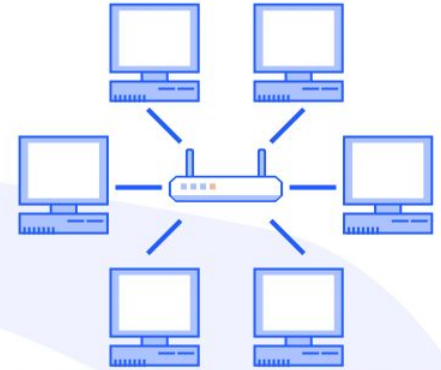
A Network of Networks



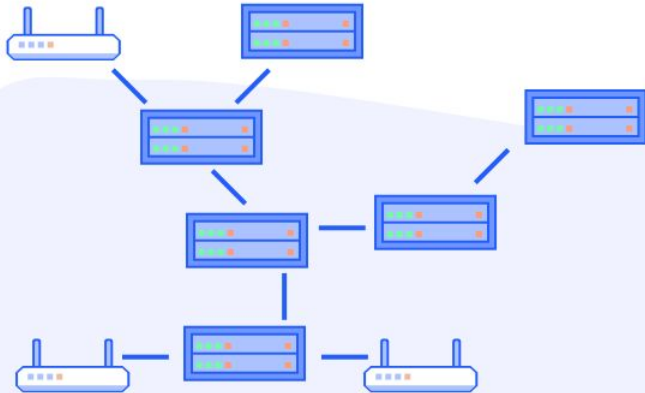
Computers can be connected directly wirelessly or with a cable.



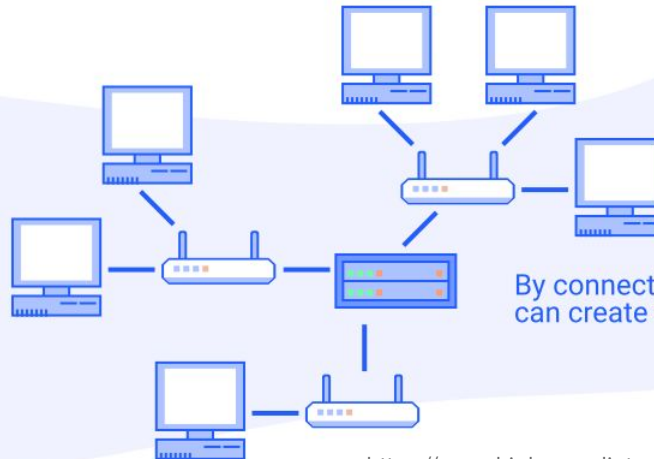
Connecting to every computer in a network becomes complicated very quickly.



Routers are specialized computers that serve as a hub, forwarding information between devices



Small networks merged into larger ones, which eventually internetworked to form one worldwide network—the Internet



By connecting routers together, we can create a network of networks



HOW THE INTERNET WORKS

SO HOW EXACTLY DOES THE INTERNET WORK? What's going on in that cloudy, tangled web? A lot of little ninja are working at super speeds to bring you the data you seek!

STEP 1 DEPARTURE

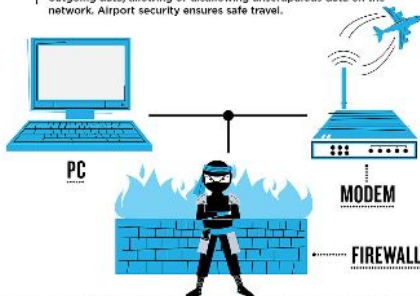
You type 'pcninja.us' into the web browser of your computer. Hop in the ninja mobile, and prepare for an adventure!

pcninja.us



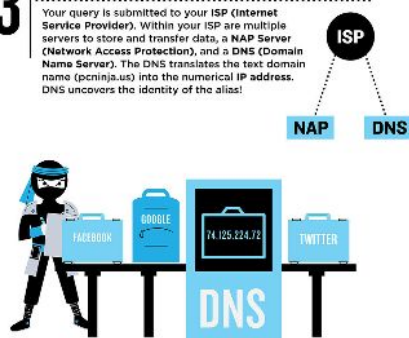
STEP 2 THE AIRPORT

Your computer is connected to the Internet through a modem and/or router, a jumping-off point to other networks in the world. Firewalls, in your browser and/or modem, monitor incoming and outgoing data, allowing or disallowing unscrupulous data on the network. Airport security ensures safe travel.



STEP 3 CUSTOMS

Your query is submitted to your ISP (Internet Service Provider). Within your ISP are multiple servers to store and transfer data, a NAP Server (Network Access Protection), and a DNS (Domain Name Server). The DNS translates the text domain name (pcninja.us) into the numerical IP address. DNS uncovers the identity of the alias!



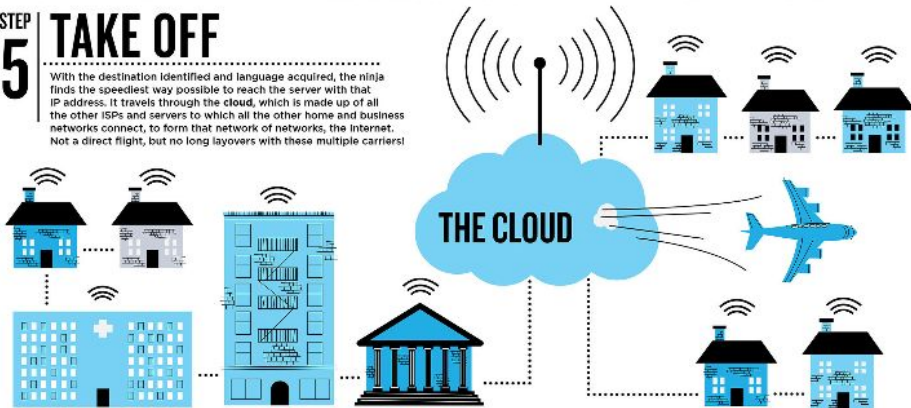
STEP 4 BUCKLE UP

Your browser maps itself to the desired IP address and establishes the hypertext transfer protocol (http), or language used to communicate on the World Wide Web. Good idea to take a pocket translator!



STEP 5 TAKE OFF

With the destination identified and language acquired, the ninja finds the speediest way possible to reach the server with that IP address. It travels through the cloud, which is made up of all the other ISPs and servers to which all the other home and business networks connect, to form that network of networks, the Internet. Not a direct flight, but no long layovers with these multiple carriers!



STEP 6 LANDING

Jumping from server to server on the Web, the ninja finally locates the target server hosting the target IP address for pcninja.us. A connection is established with that website and your computer. Please make sure your seats are in the upright position, we're ready to land!

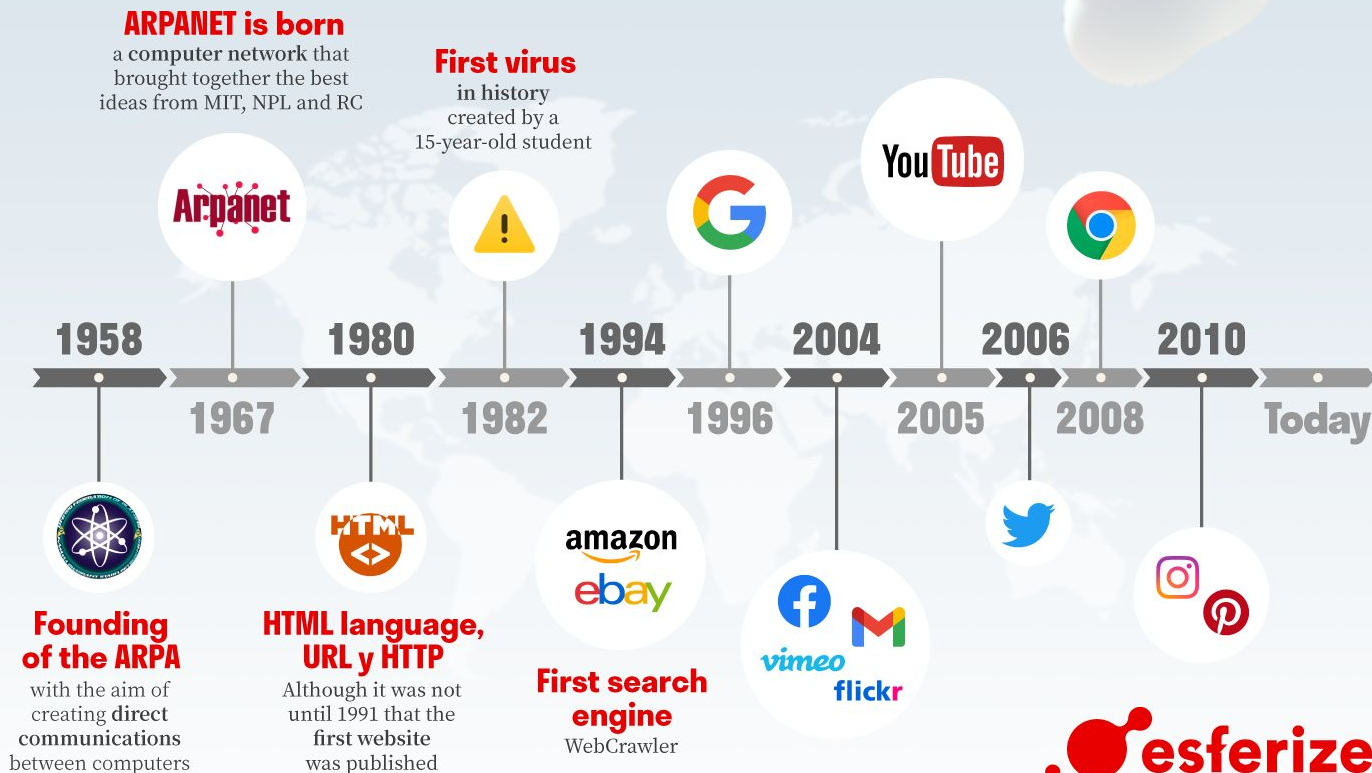


STEP 7 INCOMING!

The ninja makes an even quicker return journey, bringing to your computer screen the graphical website of pcninja.us, which is full of data, pictures, and contact information. Thank you for flying with PC Ninja, and enjoy your browse!



Evolution of the Internet



For those who are interested:

More about Internet Artifacts

<https://neal.fun/internet-artifacts>



Connect to the future

<https://www.esferize.com/en/how-has-internet-evolved/>

The Dark Web

And other fun parts of the
Internet



EXPLORING THE HIDDEN INTERNET



';--have i been pwned?

Check if your email address is in a data breach

your.email@univie.ac.at

pwned?

Oh no — pwned!

Pwned in 5 data breaches and found 1 paste (subscribe to search sensitive breaches)

    Donate

Breaches you were pwned in

A "breach" is an incident where data has been unintentionally exposed to the public. Using the 1Password password manager ensure all your passwords are strong and unique such that a breach of one service doesn't put your other services at risk.



Adobe: In October 2013, 153 million Adobe accounts were breached with each containing an internal ID, a username, email, *encrypted* password and a password hint in plain text. The password cryptography was broken and many were quickly resolved back to plain text. The unencrypted passwords were also disclosed in a public list, adding further to the risk that hundreds of millions of Adobe customers already faced.

Compromised data: Email addresses, Password hints, Passwords, Usernames



Anti Public Combo List (unverified): In December 2016, a huge list of email addresses and passwords appeared in a "combo list" referred to as "Anti Public". The list contained 950 million unique email addresses, many with multiple different passwords hacked from various online systems. The list was broadly used for "credential stuffing", that is attackers employ it in an attempt to identify other online systems where the account owners had reused their password. For detailed background on this incident, read Password reuse is a bad idea.

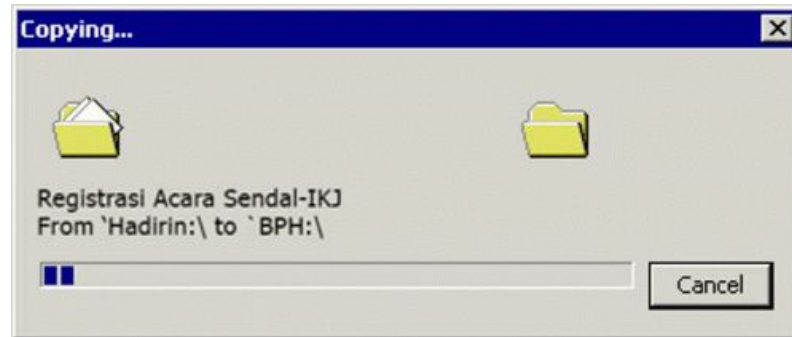


<https://haveibeenpwned.com/>

Try it yourself

How to send files online

Servers and protocols



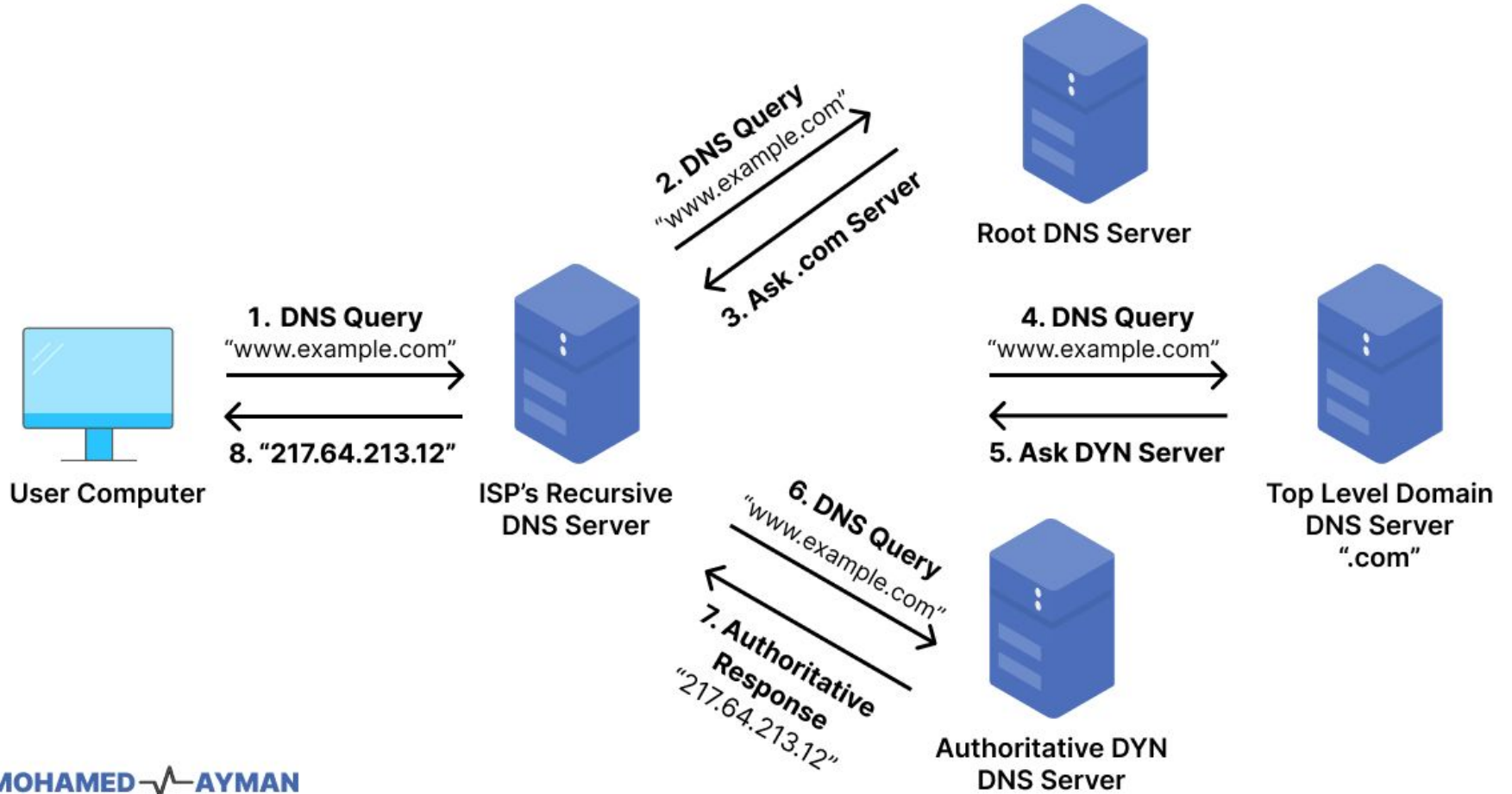
Protocol

Domain name

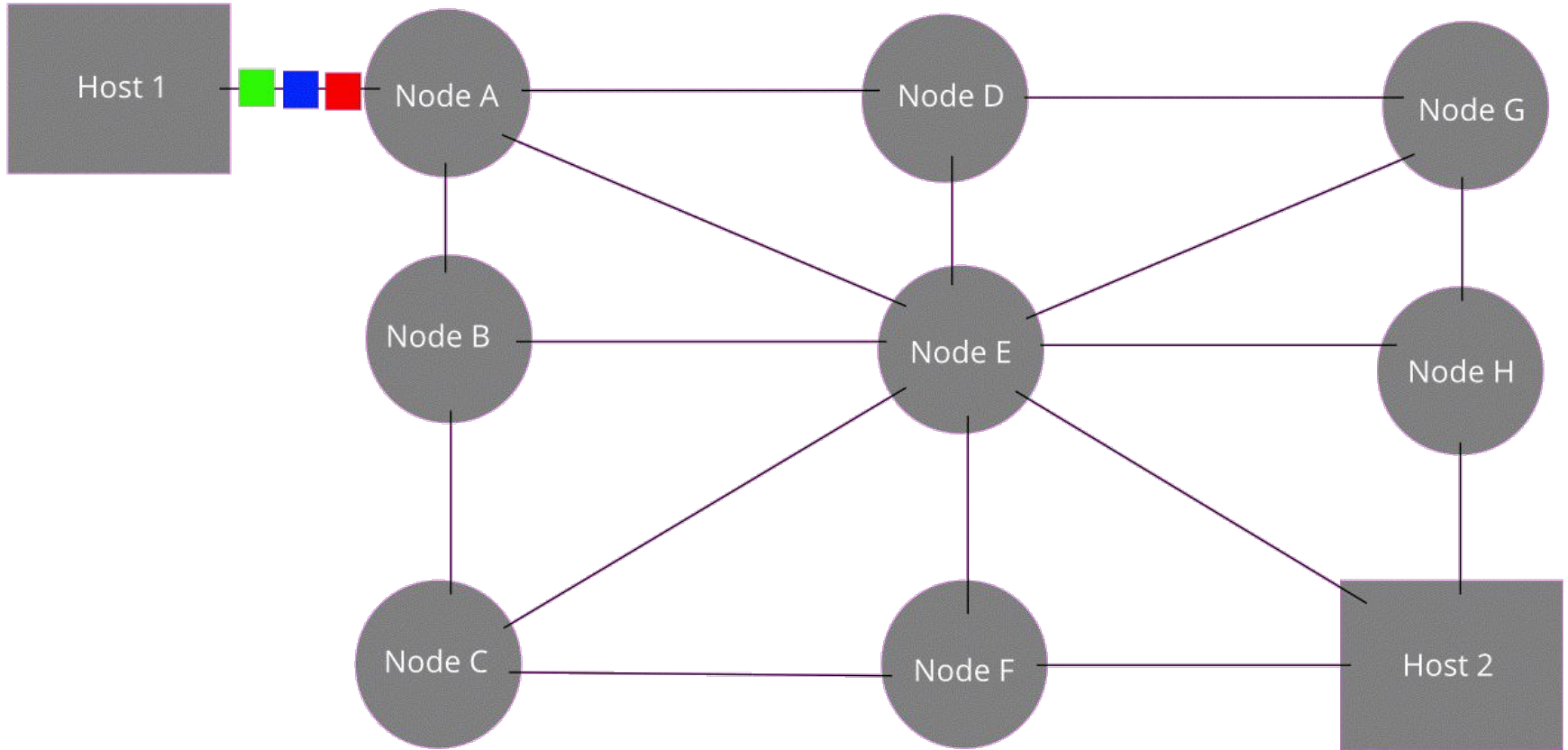
Extension

http://www.example.com/blog/what-is-a-url

URL



The original message is **Green**, **Blue**, **Red**.





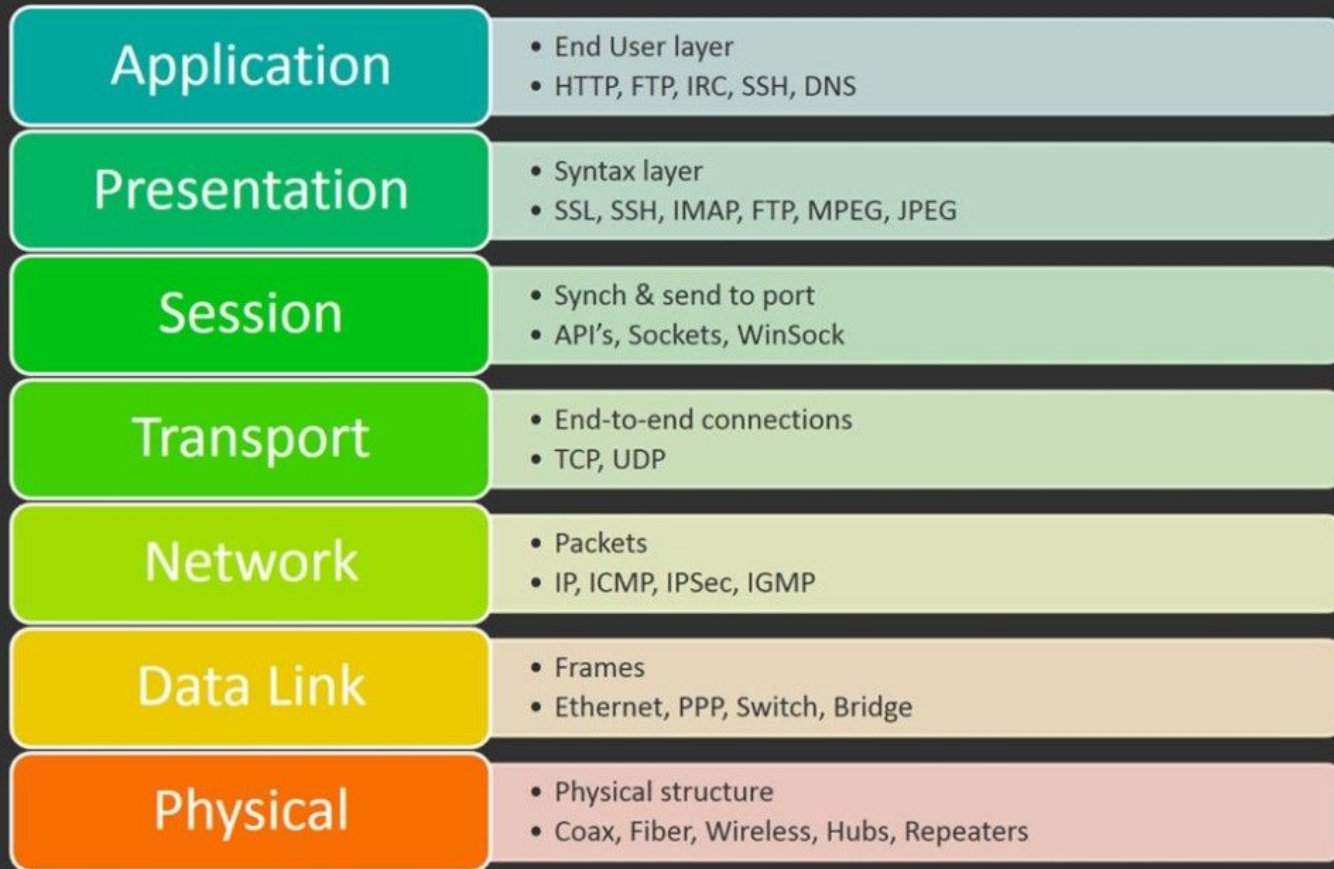
How to track your packets

The terminal command **tracert [url]** allows you to trace the “hops” from your PC to the requested url.

Open the terminal and try
tracert www.google.com
on your machine.

You can visualize the output here:
<https://stefansundin.github.io/traceroute-mapper/>

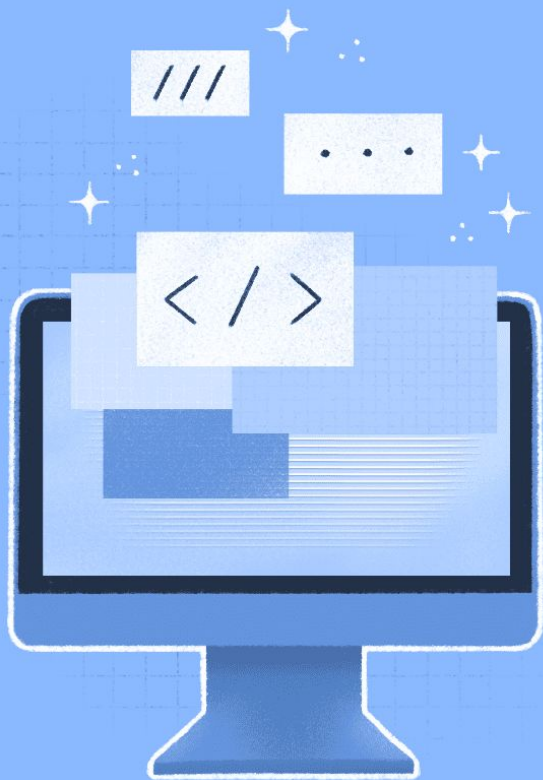
7 Layers of the OSI Model



HTML, CSS and JavaScript



Building blocks of modern websites



HyperText Markup Language (HTML)

[,āch-(,)tē-(,)em-'el]

The basic scripting language used by web browsers to render pages on the world wide web.

What's the Difference?



HTML

Hypertext Markup Language

Create the structure

- Controls the layout of the content
- Provides structure for the web page design
- The fundamental building block of any web page



CSS

Cascading Style Sheet

Stylize the website

- Applies style to the web page elements
- Targets various screen sizes to make web pages responsive
- Primarily handles the "look and feel" of a web page



Javascript

Increase interactivity

- Adds interactivity to a web page
- Handles complex functions and features
- Programmatic code which enhances functionality

HTML Page Structure

`<!DOCTYPE html>`

← Tells version of HTML

`<html>`

← HTML Root Element

`<head>`

← Used to contain page HTML metadata

`<title>Page Title</title>`

← Title of HTML page

`</head>`

`<body>`

← Hold content of HTML

`<h2>Heading Content</h2>`

← HTML heading tag

`<p>Paragraph Content</p>`

← HTML paragraph tag

`</body>`

`</html>`

Let's code our own website



Coding Session and Hands-on